SANITATION AND CLEANLINESS MANAGEMENT

The temporary and high-density nature of Simhasth will generate exceptionally high volumes of municipal waste, wastewater and sanitation demand. Waste generation will vary significantly by location and time. Food and beverage areas, parking zones, markets, accommodation areas, bathing locations, public gathering spaces and transit points may experience different waste-generation patterns.

Similarly, public toilets, urinals, bathing facilities and drinking-water points may experience highly uneven utilisation. Manual inspection and fixed collection schedules may result in either under-servicing or inefficient deployment of sanitation resources.

The challenge is to establish a data-driven, responsive and sustainable sanitation ecosystem capable of maintaining cleanliness despite rapidly changing demand.

Some of the key challenges include:

- Predicting spatial and temporal waste generation and demand for collection
- Inefficiencies of traditional collection schedules and routes.
- Segregation of waste at source
- Ensuring of cleanliness of high-footfall areas
- Managing public toilets and sanitation facilities during peak demand
- Ensuring hygiene of sanitation facilities and adequate water availability at different locations
- Piling up of food waste and other organic waste
- Rapid response to citizen/pilgrim complaints
- Ensuring sanitation operation during peak crowd periods and adverse weather

Participants are expected to exhibit capabilities in technology and/or planning. Proposals should address the aforementioned challenges and potential areas of applications can include, but are not limited to:

- Solutions that enable dynamic waste collection, routing and resource deployment for better planning.
- Monitoring of garbage bins, cleanliness of different parts of the city, citizen demands, etc. using IoT, computer vision analytics, etc.
- Predictive maintenance of sanitation infrastructure.
- Waste segregation and recycling solutions
- Sanitation performance dashboards and service-level monitoring
- Digital tracking of waste from collection to processing/disposal

This is the general direction in which applicants are encouraged to think. However, applicants are welcome to submit proposals rooted in principles that do not feature in this list but are capable of address challenges sanitation and cleanliness anticipated during the Simhasth.